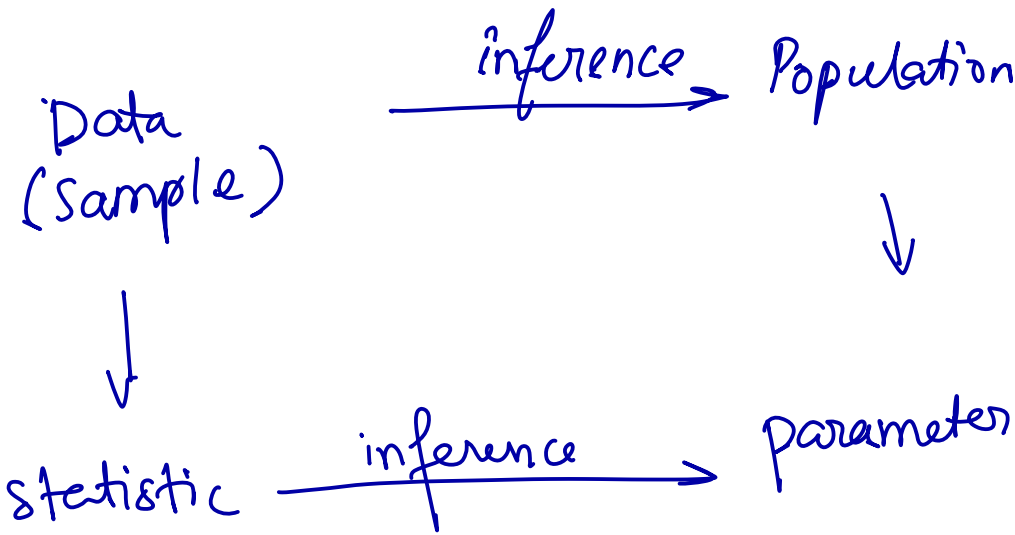
Inferential Statistics.



What is the average grade in STAT 240 ?

X = grades of students in STAT 240.

μ = average of grades of students in STAT 240.

$E(X) = \mu$.

X_1 : sample of 20 students that took STAT 240 FA 25.

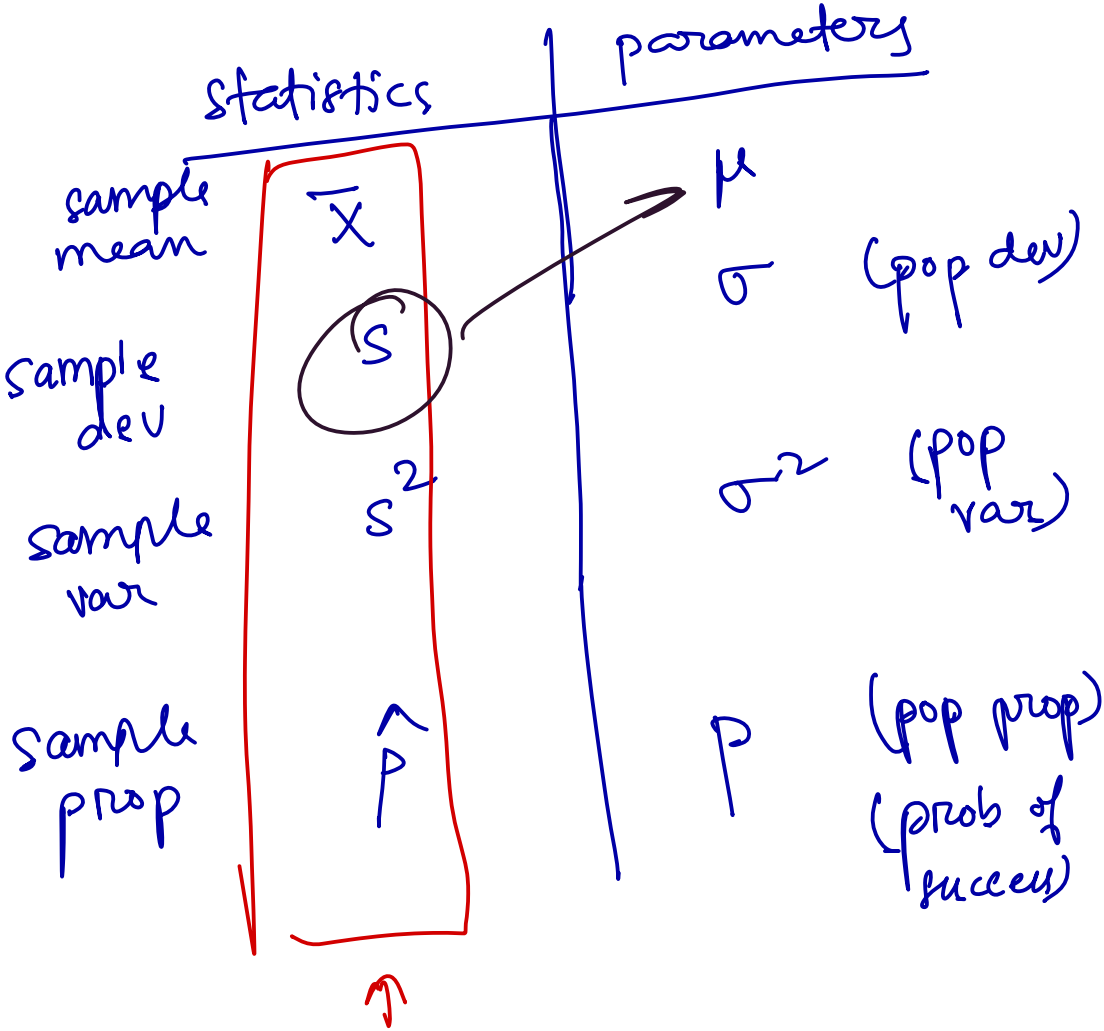
$X_1 \subset X$

↓

sample

$\bar{X}_1 \leftarrow$ avg of X_1 is going to help me understand

μ_X avg of X



estimators.

Process of finding estimator of a parameter is called

estimations.

Condition for a statistic to be an estimator.

E(statistic) = parameter

then that statistic is an estimator of that parameter.

statistic: $\bar{x}$, s, s^2, median, mode, n, $\hat{p}$.

if my parameter is μ then my estimator is $\bar{x}$.

Point estimates or point estimators.

(p.e)

$$E(\bar{X}) = \mu$$

$X \sim D(\mu, \sigma)$ — μ: avg of grades of students taking STAT 240.

σ: grades of students taking Stat 240

CLT:

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$\Rightarrow E(\bar{X}) = \mu$

$\bar{X}$

$E(\bar{X}) = \mu$.

$\bar{X}$ is a random variable

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

sampling dist

↳ sampling error or standard error.

numerical description of a sample

numerical desc of a pop

p.e ⟶ statistic. parameter.

p.e of μ ?

$\bar{X}$

$E(\text{p.e}) = \text{parameter}$.

parameter = μ

p.e = $\bar{X}$

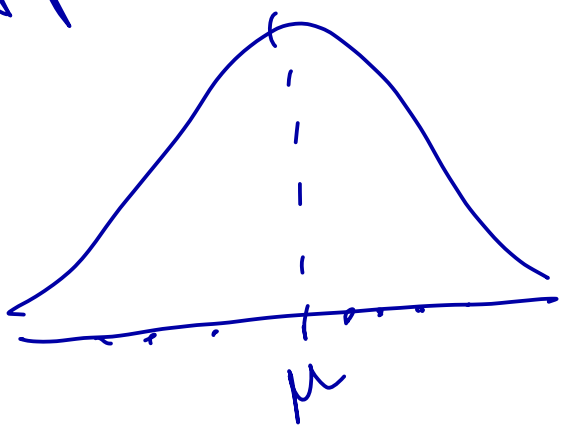
CLT : $E(\bar{X}) = \mu$ ($E(X)$)

$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

Sampling dist of p.e?

$N(\mu, \frac{\sigma}{\sqrt{n}})$

↘ dist of a p.e.



are coming from the sample

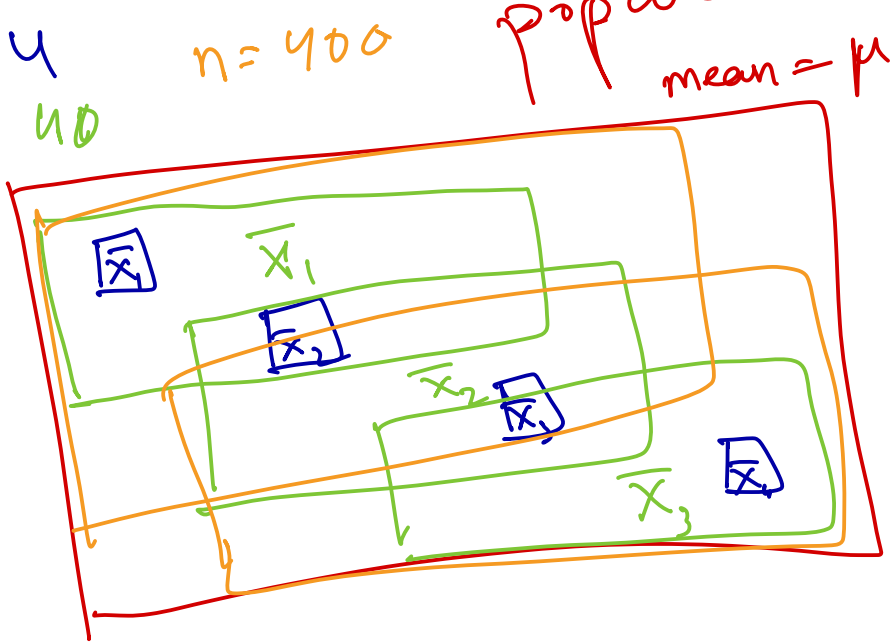
what is the name for the dev of a sampling dist?

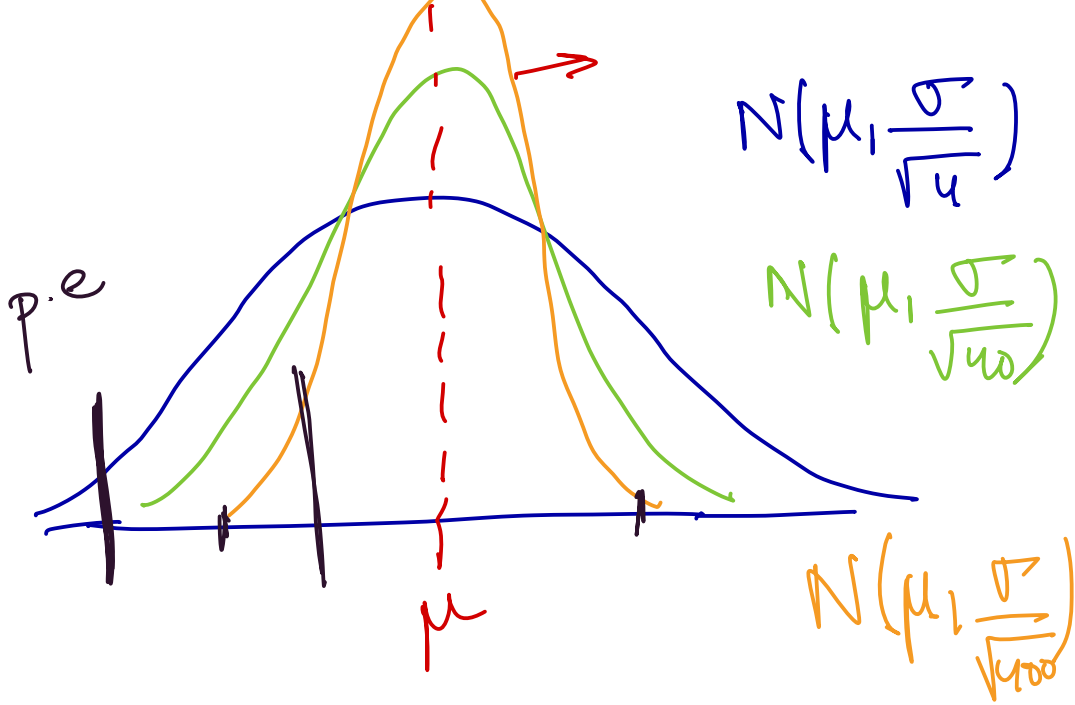
⇒ standard error (s.e) or sampling error.

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$\downarrow$ s.e $= \frac{\sigma}{\sqrt{n}}$ $\uparrow$

$n = 4$ $n = 400$ $n = 40$





s.e & sample size

μ is in here

$[a, b]$

Confidence intervals:

actual parameter is always unknown.

1) interval that contains the actual value of parameter using the sampling dist

2) confidence level (c).
(actual controls the width of the interval)

confidence ≠ prob.

confidence → If you do this experiment 100 times, then the true parameter lies inside of the interval 100c % of time.

in case of the standard normal dist bet -1 & 1 we have 68 % of the data



$\bar{x} \pm 1d$ } 68 % = 100c %.

Confidence interval:

[a, b] < sampling dist of p.e
       < conf. level.

conf. level: is not the same as prob.

General formula for CI:

mean ± z-score x dev

p.e ± C.V x S.E

m.e = margin of error.

p.e: point estimate

CV: critical value

S.E: standard error (sampling dist)

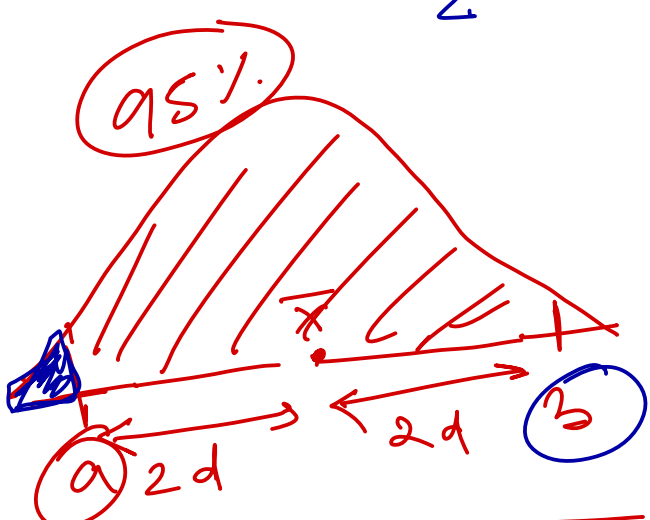
parameter: μ

p.e : $\bar{x} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

s.e : $\frac{\sigma}{\sqrt{n}}$

C.V = qdist $\left(C + \frac{1-C}{2}\right)$ ← sampling dist

$C + \frac{(1-C)}{2}$?



$\bar{x} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

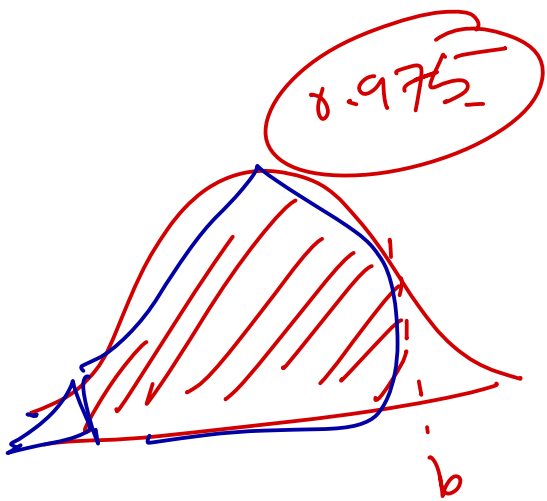
qnorm $\left(\frac{?}{0,1}\right)$

$= 2$

pnorm(2,0,1) = 0.975.

qnorm(0.975, 0, 1) = 1.9599.

↑ Z-score of point b



0.95 → 0.975

$0.95 + \frac{(1-0.95)}{2}$

$z\text{-score}_b = qnorm(0.975)$

$z = \frac{x-\mu}{\sigma} \Rightarrow x = \sigma \times z + \mu$

$-cv$: s.e, p.e

$x = \sigma(-z) + \mu$

σ : s.e, z : cv, μ : p.e

b

$(p.e - c.v \times s.e,\ p.e + c.v \times s.e)$

Hypothesis testing :

hypotheis :

H_0 : null hypotheis (universal truth)

H_α : alternative hypo (your test)

H_0 is true.

H_0: true parameter $\leq$ / $=$ / $\geq$ value

H_a: true parameter $\geq$ / $\neq$ / $<$ value



H_0 : $=$ (simple hypo testing)

H_0 : $\leq$ / $\geq$ (composite hypo testing)

direction of the test is decided on the direction of H_a :

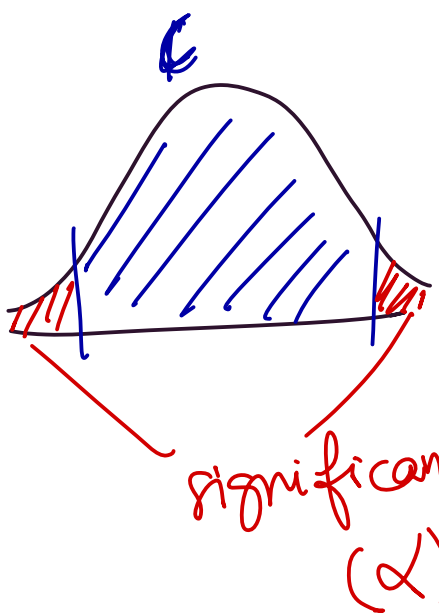
$H_a: <$ Left tailed test

$H_a: >$ Right tailed test

$H_a: \neq$ Two tailed test.

Sample → p.e

E(p.e) = parameter.



1) Gathering evidence:

→ Coming from the sample, we will need the p.e.

p.e ~ sampling dist.

$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$

$\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$

standardize

test statistic = $\frac{\text{p.e} - \text{parameter}_{H_0}}{\text{s.e}}$ ($\text{parameter}_{H_0} \approx E(\text{p.e})$)

~ null dist

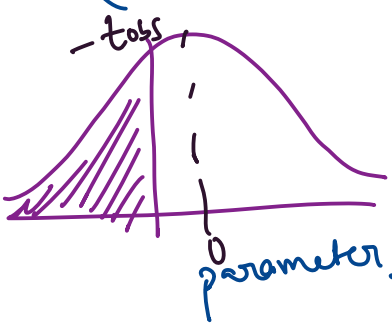
test statistic's dist is called null dist. (standardize version of the sampling dist).

Prove that the evidence either supports or doesn't support H_0?

P-value = prob of getting the observed test statistic under the H_0 as or more extreme.

Left tailed

$P(t \leq t_{obs})$



Right tailed

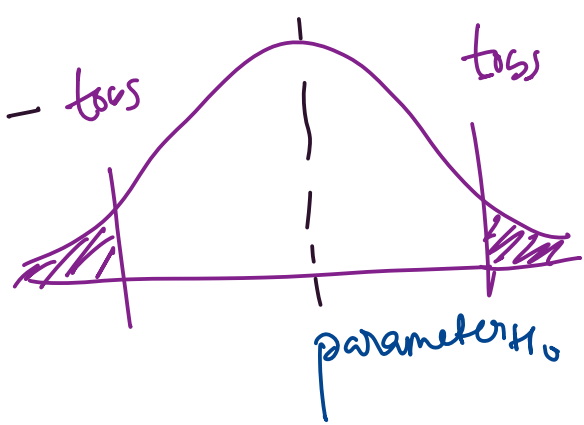
$P(t \geq t_{obs})$



two tailed

p-value =

$2 \times P(t \geq |t_{obs}|)$



general formula for p-val:

$p\text{-val} = P(t > |t_{obs}|)$

Hypothesis testing:

1) H_0, H_α & checking the model & assumptions.

2) test statistic. (standardized version of the sampling dist)

collecting evidence from the sample

$$= \frac{p.e - \text{parameter}_{H_0}}{s.e(p.e)} \quad p.e$$

$$\left(\frac{x - E(x)}{s.d(x)} \right)$$

$\sim$ null dist (standardized version of sampling dist)

↓

assuming H_0 is true.

3) Calculate the prob of getting our obs test statistic.

p-val = Prob of getting a ts as or more extreme than the one observed

↓

gets decided by H_α.

$H_\alpha: <$

$P(ts \leq ts_{obs})$

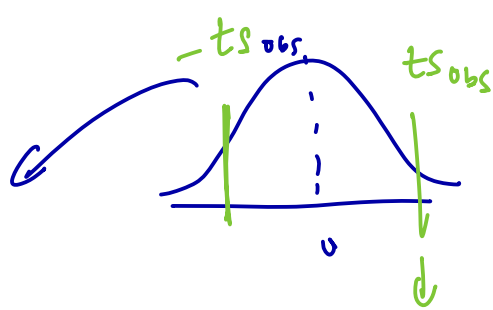
$H_\alpha: >$

$P(ts \geq ts_{obs})$

$H_\alpha: \neq$

p-val

$2 \times P(ts \leq -ts_{obs})$



$2 \times P(ts \geq ts_{obs})$

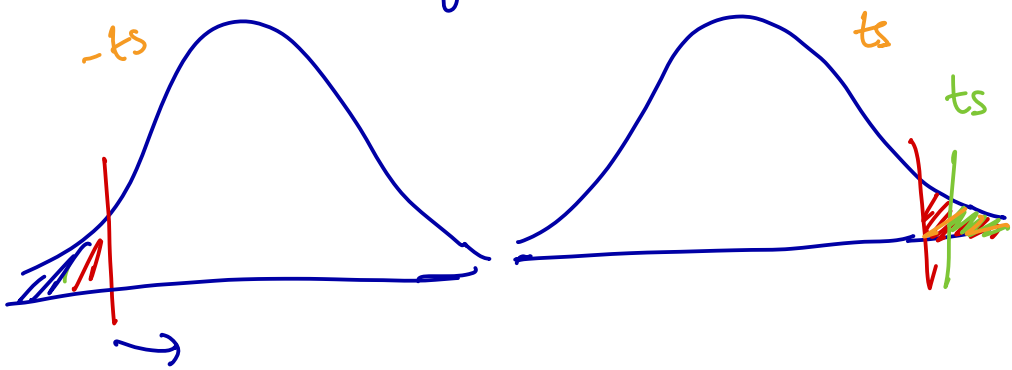
4) Conclusion

significance level

α (= 1 − c)

p-val $<$ α

$\Rightarrow$ Reject H_0.



$\rightarrow$ p-val $\geq$ α.

$\Rightarrow$ Fail to reject H_0.

5% significance level

or $\alpha = 0.05$.

$\alpha : 0.05 \quad c = 0.95$

Reject H_0. $[a, b]$

H_0 is not in $[a, b]$.

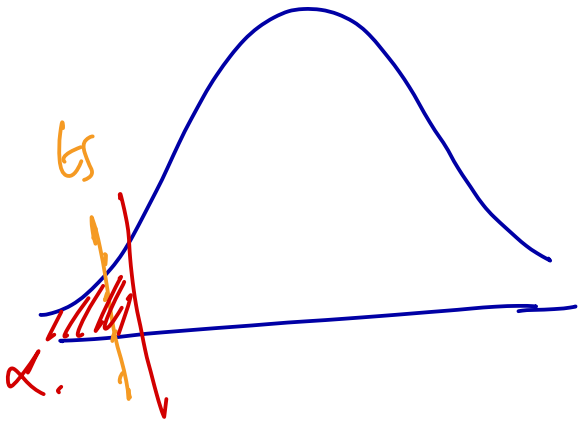
Fail to reject H_0, $[a, b]$

$H_0 \in [a, b]$.

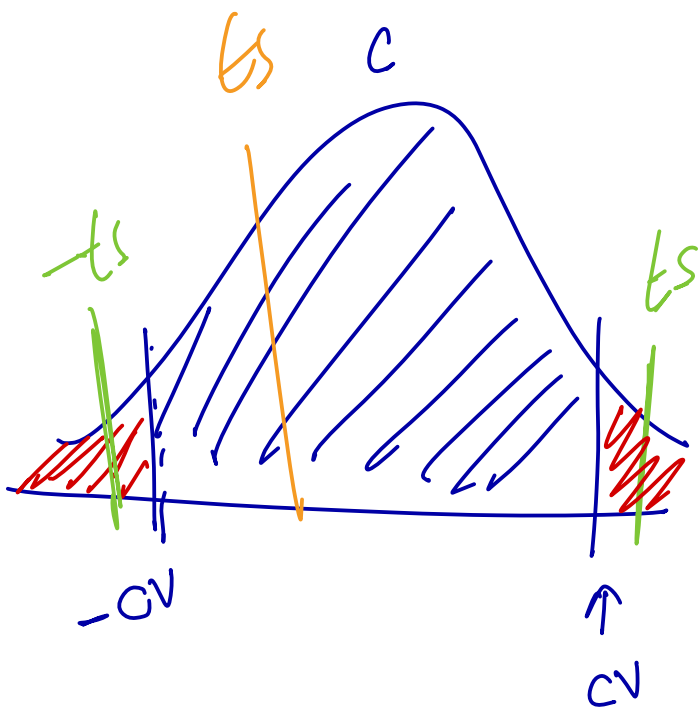
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p-val = 0.04995.

0.05001



p.e ± c.v x s.e



p.e with cv.